CardioClarity Studio

Live demo	cardioclarity-962866578386.us-central1.run.app
Code	github.com/ashishmokalkar/Multimodal-AI-Hackathon-Devpost
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MULTIMODAL AI HACKATHON 2026 · TRACK A · PROJECT DOCUMENTATION

Coronary risk you can see on the heart, with a reason behind every number

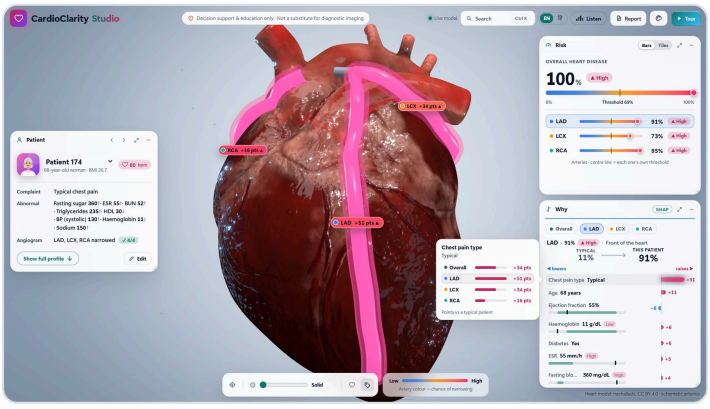
The problem. Heart disease is India's biggest killer: **2.8 million deaths a year, 28% of all deaths** (GBD India, *Lancet Global Health* 2018). It strikes Indians about a decade earlier than Western populations, and **52% of these deaths come before the age of 70**, against 23% in the West (Prabhakaran et al., *Circulation* 2016): parents and earners in their working years. India has **fewer than five cardiologists per million people**, and five states hold 79% of its cath-lab capacity (*JSCAI* 2026). So most patients get **about two minutes** with a primary-care doctor (Irving et al., *BMJ Open* 2017), are handed reports in English, which only about **one in ten Indians** speaks (Census of India 2011), and leave with at most a single risk number: not **where** the disease is, not **why**, and not **what to do next**.

Our solution. CardioClarity Studio turns the tests a patient already has (blood tests, ECG, echo) into an answer a doctor can read at a glance and a patient can understand: a **simple dashboard** with the risk of coronary artery disease (CAD) and of narrowing in each main artery (LAD, LCX, RCA) shown on the heart itself, the reasons in plain words, and **clear next steps** (the tests to ask about and when to seek help at once), spoken in **English or Hindi** through Sarvam AI, with more Indian languages designed. With consent, records are designed to arrive through ABHA, India's digital health ID (93.95 crore accounts by July 2026), instead of being typed in.

00 How CardioClarity meets the Track A evaluation criteria

CRITERION	WHAT WE DELIVER	EVIDENCE TO CHECK
Predictive performance 30% · pages 2–3	Best model per target from 10+ families, TabPFN and AutoGluon. **Validation:** 10 × 5 repeated stratified CV, half for every choice, half opened once at the end. Calibrated, with conformal uncertainty.	**ROC-AUC: CAD 0.939, LAD 0.869, LCX 0.782, RCA 0.770**, on held-out repeats, with 95% CIs.
3D visualization 25% · page 4	Anatomical glTF heart; LAD, LCX and RCA as separate objects. **Spatial risk mapping:** colour, thickness, glow and flow per artery. Click, fly-to, X-ray, live transitions.	**56–59 fps on integrated Intel graphics, no dedicated GPU** (measured).
Clinical interpretability 20% · page 5	**Feature attribution with SHAP** per patient and target. Breakdown against normal ranges, what-if, plain-language report, spoken summary in English and **Hindi (Sarvam AI)**.	Press “Explain this prediction”; SHAP points add up exactly to the score.
System integration 15% · page 6	One typed contract, model to browser. **Real-time updates:** a what-if re-scores, re-explains and recolours the heart. ABHA, report-scan and voice intake designed on it.	**About 1.0 s** from slider to recoloured heart, live (measured).
Technical implementation 10% · page 6	FastAPI + React, one Docker image on Cloud Run (CPU only). **Reproducibility:** one command rebuilds the models.	70 Python and 163 frontend tests pass; 0 accessibility violations (axe).

Decision support and education only. Not a substitute for diagnostic imaging. Shown on every screen and report page.



Patient 174, 68: all three arteries narrowed on her angiogram, and the model matched all four targets. Why explains the LAD result (91%) input by input; hovering “Chest pain type” shows its points for every target, on the heart too.

01 Predictive performance: data, preprocessing, validation 30%

We use the **Extension of the Z-Alizadeh Sani dataset** (UCI #411): 303 patients from one Iranian heart centre, all of whom had coronary angiography, with 55 inputs and four labels. Narrowing of 50% or more counts as disease.

LABEL	POSITIVE	SHARE	TERRITORY
CAD	216	71%	Any coronary artery disease
LAD	177	58%	Front of the heart
LCX	119	39%	Side and back
RCA	114	38%	Right side and bottom

Ages 30–86 (mean 58.9); 127 women. Of 216 CAD patients, 86 have one artery narrowed, 67 two and 63 all three. Because always answering "CAD" is 71% accurate, we lead with ROC-AUC, balanced accuracy and F1.

Input groups (49 served inputs)

`Demographics`	Age, sex, weight, height (BMI derived)
`History`	Diabetes, hypertension, high cholesterol, smoking, family history, kidney failure, stroke, heart failure, airway and thyroid disease
`Exam`	BP, pulse, oedema, weak pulses, lung crackles, murmurs
`Symptoms`	Chest pain type, breathlessness, functional class, low-threshold angina
`ECG`	Q waves, ST elevation and depression, T inversion, LVH, poor R progression, bundle branch block
`Labs`	Fasting sugar, creatinine, lipids, BUN, ESR, haemoglobin, electrolytes, blood count
`Echo`	Ejection fraction, regional wall-motion abnormality, valve disease

Data cleaning

No missing values and no duplicate rows, so the work was consistency and encoding. The loader rejects any unexpected value instead of filling it in.

- **No label leakage:** LAD, LCX, RCA and Cath are labels only, never inputs, as the brief requires; the API rejects them as inputs too.
- **Constant column dropped:** "Exertional chest pain" is "No" for every patient.
- **Redundant columns:** BMI equals weight / height² exactly and "Obesity" is BMI ≥ 25 (one exception), so only BMI is kept.
- **Merged codes:** the mutually exclusive typical, atypical and non-anginal flags become one four-level chest-pain type; current and ex-smoker become one smoking status.
- **Row order:** prevalence falls from 80% in the first rows to 33% in the last, so every split is shuffled and stratified.

Feature engineering

A feature was kept only if it raised the average AUC across three untuned models and four targets, and no model lost more than one corrected standard error.

- **Pre-test probability** (Diamond–Forrester, from age, sex and pain type): the strongest single feature, univariate AUC 0.83 for CAD.
- **Ten published clinical risk scores** (ESC 2019 and 2024, CAD Consortium, Duke, Jang and others), kept where they helped: Duke for LAD, all ten for LCX, Jang for RCA.
- **Wall-motion flags** for LAD: abnormal motion in three or more regions meant LAD disease in 28 of 28 patients.
- **Comorbidity count** replaces eight rare yes/no columns.
- **Per-target feature selection** inside each training fold: 45 inputs for CAD, 53 for LAD, 66 for LCX and 44 for RCA.
- **Linear-model preprocessing** per fold: 1st/99th-percentile clipping and log transform of skewed labs.

Validation methodology

- **Fixed folds:** 10 repeats × 5 stratified folds (stratified on the CAD and artery pattern), saved once and reused by every experiment, so all comparisons are paired.
- **Choose on repeats 0–4, report on repeats 5–9.** Cleaning, features, tuning, model choice, ensemble weights and calibration used repeats 0–4 only. Repeats 5–9 were opened once, for the final numbers.
- **Nested tuning:** Optuna searches inside the training folds of repeats 0–2.
- **Paired significance:** a difference counts only if it exceeds one Nadeau–Bengio corrected standard error; 95% CIs from a patient bootstrap.
- **Learned inside the fold:** scaling, clipping, feature selection, calibration, thresholds and conformal sets are fitted on training patients only. Nothing is fitted on all 303 before splitting.

What we searched

Logistic regression, SVM, kNN, naive Bayes, random forest, extra trees, LightGBM, XGBoost, CatBoost and EBM, all re-tuned; TabPFN v2.5–v3.5 with tuning and fine-tuning; 27 AutoGluon model families with stacking and bagging; joint, hierarchical and chained multi-label models; pooling with an external cohort (AngioCAD, 377 patients). Each target got its own winner by mean AUC on repeats 0–4.

Exploratory findings

- **Chest pain type and age** lead for every target.
- **LAD shows on echo:** low ejection fraction and wall-motion abnormality (the LAD supplies the front wall).
- **RCA carries an inflammation signal** (neutrophils and ESR up); **LCX** follows age, blood pressure and metabolic factors and is silent on the ECG.

01 Predictive performance: model architecture and evaluation results 30%

The four models (one per target)

TARGET	MODEL	INPUTS
CAD	Explainable Boosting Machine (glass-box additive model)	45
LAD	CatBoost, re-tuned, with the Duke score	53
LCX	XGBoost, re-tuned, with all ten risk scores	66
RCA	Logistic stack of 8 models (LightGBM, logistic regression, EBM, kNN, naive Bayes, random forest, SVM, extra trees)	44

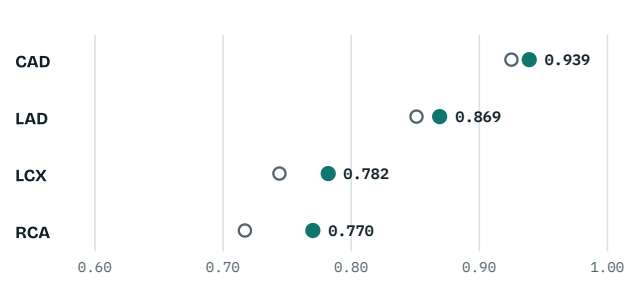
Risk estimation after the model

- **Calibration:** Platt scaling, fitted inside each training fold.
- **Consistency:** an artery's risk is capped at the CAD risk and flagged only when CAD is, so the heart never shows an artery riskier than any disease at all.
- **Thresholds:** per target, for the best balanced accuracy; lower for LCX and RCA, favouring catching disease.
- **Uncertainty:** a 90% split-conformal set; when it holds both answers the app marks the result “unsure”.

Evaluation results of the served pipeline on the held-out repeats (25 test folds)

TARGET	ROC-AUC (95% CI)	ACCURACY	BALANCED ACC.	PRECISION	RECALL	F1	BRIER	ECE
CAD	**0.939** (0.903–0.960)	0.867	0.868	0.944	0.867	0.903	0.088	0.040
LAD	**0.869** (0.822–0.906)	0.792	0.780	0.807	0.850	0.826	0.144	0.054
LCX	**0.782** (0.722–0.820)	0.673	0.698	0.561	0.812	0.660	0.191	0.052
RCA	**0.770** (0.714–0.813)	0.679	0.678	0.573	0.674	0.612	0.192	0.056

Exactly what the app serves (calibration, cap, artery rule and thresholds), with all post-processing fitted per outer fold on training patients. ECE: expected calibration error (lower is better).



The served pipeline improves on the best untuned model from exploration by 0.014 (CAD), 0.018 (LAD), 0.038 (LCX) and 0.053 (RCA). In model selection, the gains over an untuned logistic regression on the raw inputs were significant for LAD (+0.055), LCX (+0.121) and RCA (+0.054); for CAD the untuned baseline already reaches 0.925.

Not kept: TabPFN (best untuned, but never better than the tuned models), AutoGluon stacks (no gain; worse for RCA), multi-label strategies, and pooling AngioCAD (cross-cohort AUC 0.61–0.78).

Calibration, before → after Platt scaling

TARGET	ECE	SLOPE
CAD	0.044 → 0.040	1.01 → 0.97
LAD	0.075 → 0.047	0.70 → 0.94
LCX	0.061 → 0.053	1.20 → 0.92
RCA	0.138 → 0.047	0.40 → 0.92

“70%” should be right about 70% of the time. Temperature, beta and isotonic calibration were also compared.

Strong and weak groups (ROC-AUC)

GROUP (N)	CAD	LAD	LCX	RCA
Men (176)	0.908	0.876	0.776	0.725
Women (127)	0.965	0.844	0.746	0.805
Age 50–64 (150)	0.946	0.867	0.780	0.770
Age 65+ (95)	0.848	0.807	0.651	0.677
Diabetic (90)	0.975	0.739	0.727	0.694

Weakest for patients 65 and over and for LAD in diabetics; the app warns on screen for patients 65+.

Study patients are never scored by a model trained on them

A model trained on all 303 patients scores them almost perfectly (AUC 0.99–1.00) because it has memorised their labels. So each target is served as **five cross-fitted fold models**: a study patient, or a lightly edited what-if of one, is scored by the fold model that never saw them; a new patient gets the average of all five. Scored this way through the live API, the app agrees with the angiogram for **87%** of patients on CAD, **81%** on LAD and **71%** on LCX and RCA, in line with cross-validation.

Uncertainty and artery localisation

The 90% conformal sets hold their coverage (0.889–0.900). For CAD the model commits to one answer for 96% of patients; for LCX and RCA it cannot decide for about half, and shows those as “unsure”. Among CAD patients, picking the involved artery is only slightly better than chance (AUC 0.55–0.62), so the heart shows **likelihood of involvement**, never a lesion location. Refitting with new seeds moves AUC by at most 0.004.

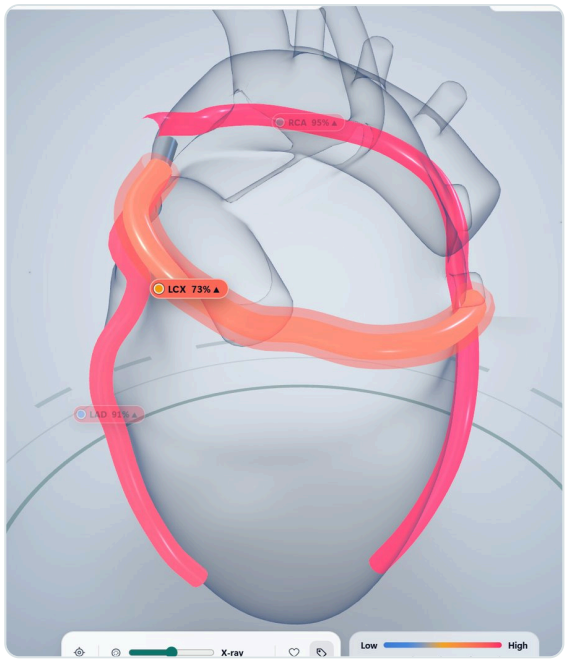
02 3D visualization: anatomy, spatial risk mapping and interaction 25%

Anatomical representation

- **Heart:** the CC BY 4.0 “Realistic Human Heart” glTF mesh by neshallads (about 22,500 faces), loaded once and reused for the live view and the printed report.
- **Arteries:** LAD, LCX and RCA (plus the left main stem) are our own schematic tubes, Catmull–Rom curves laid over the heart's surface in their anatomical territories. Each is a separately addressable object. They show which territory a prediction refers to, not the patient's own vessels.

Spatial risk mapping

- **One colour scale per artery, centred on its own threshold:** blue (low) → amber (at the threshold) → red (high). The same scale is printed under the heart and used in every window, so amber always marks the threshold.
- **Risk drives four channels at once:** colour, tube thickness, glow and blood-flow speed (instanced particles). Flagged arteries pulse with the heartbeat, which beats at the patient's own pulse rate.
- **Labels on the heart** follow each artery and show its percentage, ▲ when flagged and “?” when the model is unsure. While an input is hovered in Why they switch to that input's points for each artery, and arteries it barely moves fade.
- **Live transitions:** when a what-if changes the score, every channel eases over 600 ms, so the change in risk is visible on the artery itself.



X-ray mode, LCX selected (patient 174, 73%). Clicking an artery dims the others, flies the camera to it and switches the Why window to that artery.

Interactive visualization

Explore
Orbit and zoom; a depth slider from solid heart through X-ray to arteries only; fly to any artery and back to the overview; labels on or off.

Link
Click an artery on the heart or in the Risk window to select and explain it; hover an input to see its effect on all three arteries at once.

Guide
A guided tour flies through the overall result and each artery. Ctrl+K finds any patient, input, view or window.

From prediction to pixels

API	ARTERY MAP	SCALE	REACT THREE FIBER	REPORT
One response	**Look up**	**Map**	**Render**	**Snapshot**
CAD and per-artery probability, threshold, unsure flag and Shapley points.	Each target id (LAD, LCX, RCA) maps to exactly one tube, colour and label.	Probability → colour, thickness, glow and flow, relative to that artery's threshold.	Three.js scene, bloom, eased transitions; frames only when needed if motion is off.	The same scene renders still images for the PDF: the full heart and each artery lit.

Consistent LAD, LCX and RCA correspondence

The heart, the Risk window and the Why window all read the same response object through one artery map, so the colour of an artery and the number beside it can never disagree. Arteries are keyed by the API's target ids, not by position, and the frontend validates every response against the typed contract before anything is drawn.

Accessible 3D: the canvas carries a text summary of all four results for screen readers; reduced-motion settings stop the heartbeat, flow and rotation.

Runs without a dedicated GPU (measured)

GRAPHICS	ANIMATING	ORBITING	95% OF FRAMES
Intel UHD integrated	**59 fps**	**56 fps**	≤ 18.5 ms
Software only (SwiftShader, CPU)	9 fps	9 fps	≤ 200 ms

Microsoft Edge, 1440 × 900 window, patient 151, heartbeat and blood flow on; frames counted over 8 s. A performance monitor drops to a lighter render setting if frames fall. With no graphics hardware at all, the scene still renders, at about 9 fps.

03 Clinical interpretability: SHAP explainability, breakdown and what-if 20%

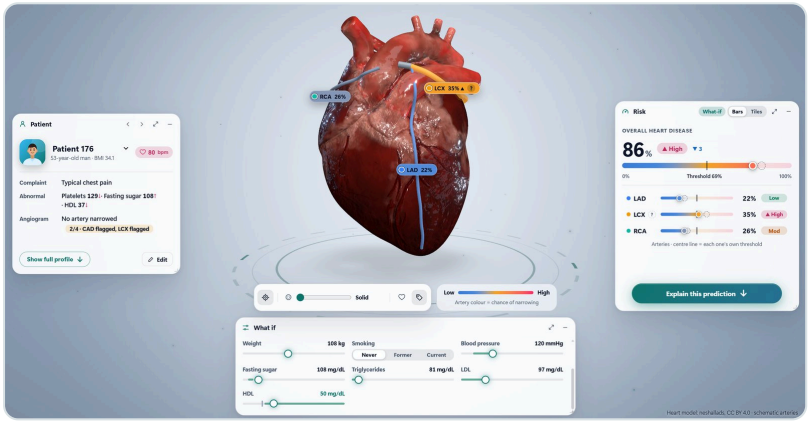
Feature attribution quality (SHAP)

Every response carries **SHAP values** (Shapley additive explanations) for all four targets, computed with the Baseline Shapley method (32 orderings, against a typical reference patient) on the probability the app actually shows. They add up exactly to this patient's probability minus the typical patient's, so “+51 pts” on the heart means the same as “+51” in the Why window.

We run SHAP on the served probability rather than a model-specific explainer, because the four models differ (a glass-box model, two tree ensembles and a stack): one method gives the same units everywhere and includes calibration and the cap. We chose SHAP over LIME because SHAP values add up exactly and do not change between runs.

Physiological factor breakdown

- **Why window:** a journey from the typical patient's risk to this patient's; each row shows the input, the value with its unit, a normal-range bar, a High/Low flag and signed points, largest first. Click an artery to switch every row to it.
- **Factor map:** the strongest inputs against all four results at once.
- **Measures:** every input as a tile against its typical range, filterable to what needs attention; labs switch between mg/dL and mmol/L.
- **Context:** risk among same-age, same-sex study patients, and what the angiogram found.



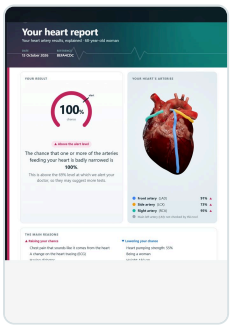
What-if, patient 176 (53, smoker, clear angiogram). Setting smoking to “never” and HDL from 37 to 50 mg/dL re-scores live: LCX 51% → 35% (▼16), LAD 28% → 22%, CAD 89% → 86%. Dotted markers keep the original values in view.

What-if, in real time

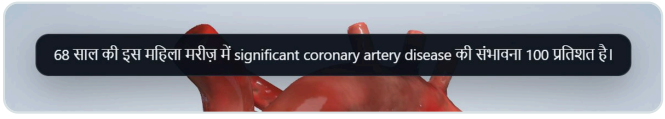
Sliders for the inputs a patient can change (weight, smoking, blood pressure, fasting sugar, triglycerides, LDL, HDL) send a separate request to the same API. The Risk window, Why window and heart show the scenario next to the original, so nothing about the real result is overwritten. A clinician can show a patient what the model would expect if, for example, they stopped smoking.

A what-if shows what the model would predict, not what treatment would achieve.

Clarity of risk explanation for the patient



- **A 3-page report** (PDF): the chance with the 3D heart and the main reasons; each artery lit on the heart; take-home numbers, questions for the doctor and when to get help.
- **Listen:** a captioned spoken summary of the chance, the arteries, the four main factors, out-of-range readings and a next step.



Listen in Hindi. An EN | हिं switch beside Listen; the Hindi summaries are recorded with **Sarvam AI** (Bulbul v3) and keep the medical terms in English, as Indian clinicians speak.

Decluttered by design

Clinical software often shows every value at once. The interface shows the answer first and detail on request.

- **The heart is the interface.** Results live on the anatomy, with small floating windows around it instead of a dashboard of charts.
- **Two windows by default** (Patient and Risk). Why opens from “Explain this prediction”; the factor map, measures, comparison and model card open on demand (Ctrl+K).
- **The patient summary lists only abnormal values;** the full 49-input profile is one click away.
- **Start screen:** four featured patients that each tell a different story, search across all 303, and adding a new patient beside them.
- **One disclaimer line** on every screen instead of pop-ups; plain words in the report, no jargon. Two themes, full keyboard use.

The four featured patients against their angiograms

PATIENT	STORY	CAD	LAD	LCX	RCA	ANGIOGRAM	MATCHED
174 · 68 F	Typical chest pain, diabetes	100%	91%	73%	95%	LAD, LCX, RCA narrowed	4/4
125 · 36 F	Breathless, no chest pain; caught from echo and ECG	85%	75%	14%	12%	LAD narrowed	4/4
151 · 42 M	Typical pain, smoker, family history	29%	28%	29%	17%	Clear	4/4
176 · 53 M	Typical pain in a smoker; the model over-calls	89%	28%	51%?	29%	Clear	2/4

Each scored by the fold model that never saw them; “?” marks unsure. For 176, Why shows the over-call comes from typical chest pain.

04 System integration: data pipeline and real-time updates 15%

BROWSER
Inputs
React sends the 49 values to `POST /predict`, checked against the feature contract first.

FASTAPI
Score
Five fold models per target; the one that never saw a matching study patient scores it, else their average.

FASTAPI
Post-process
Platt calibration, artery cap, threshold, conformal “unsure”.

FASTAPI
Explain
SHAP values (Baseline Shapley) for all four targets, 32 orderings.

BROWSER
Show
Heart, Risk, Why and Patient update together from one response.

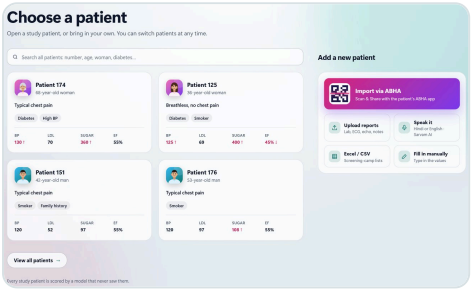
Real-time visualization updates (measured)

Live: the Cloud Run deployment; local: this laptop. Edge, patient 151, weight slider.

SLIDER → RECOLOURED HEART	LIVE	LOCAL
Pause while the slider moves	300 ms	300 ms
API round trip: 4 models + explanations	671 ms	505 ms
End to end, median of 10	**1.00 s**	**0.84 s**

One contract, end to end

The API contract (v1.1.0) is typed on both sides: Pydantic in Python and Zod in TypeScript, so a mismatch fails loudly instead of showing a wrong number. Inputs are listed by `GET /features` with units, ranges and options, and the frontend builds its forms, sliders and tiles from that list. Values outside the training range raise a visible warning.



Start screen: featured patients and search, and five ways to add a patient.

Patient data intake, designed for India

Entering 49 values by hand is slow, and most already exist in the patient's records. Every route below ends at one review screen, so nothing is scored until a person confirms the values.

- `LIVE` **Fill in manually**, with units, ranges and validation from the contract.
- `DESIGNED` **Import via ABHA:** the patient scans the clinic's QR code with their ABHA app (Scan & Share) and approves consent; labs, diagnoses and vitals arrive as FHIR R4 records through the Ayushman Bharat Digital Mission and are mapped onto our 49 inputs.
- `DESIGNED` **Upload reports:** photos, PDFs or WhatsApp forwards of lab, ECG, echo and clinic reports, read by a vision model that shows where each value came from.
- `DESIGNED` **Speak it** in Hindi or another Indian language, transcribed by Sarvam AI speech-to-text and turned into values.
- `DESIGNED` **Excel / CSV** for screening camps: score many patients and rank who to refer first.

05 Technical implementation: architecture, code quality, reproducibility 10%

- **Architecture:** `ml/` (cleaning, features, CV, model search, export), `models/` (cross-fitted models and `manifest.json`), `api/` (FastAPI), `frontend/` (React 18, TypeScript, React Three Fiber, Zustand, TanStack Query). One Docker image on Cloud Run, 2 vCPU, CPU only.
- **Extensible without a redesign:** new inputs come through `/features`, new models through the manifest, new vessels through the artery map.
- **Code quality:** 70 Python tests (data, folds, features, export, API contract, including that study patients are scored by models that never saw them); 163 frontend tests; strict TypeScript and ESLint; automated browser checks at several screen sizes in both themes.
- **Usage:** open the live demo, pick a featured patient, press “Explain this prediction”, click an artery, try What-if, open Report or Listen. The Tour walks through everything.

```
# Dashboard and API in one container
docker build -t cardioclarity .
docker run -p 8080:8080 cardioclarity

# Or separately (Python 3.11, Node 24)
pip install -r requirements.txt
uvicorn api.main:app --port 8000
cd frontend && corepack pnpm install
corepack pnpm dev

# Reproduce: data and the served models (CPU)
python scripts/download_data.py
python -m ml.export
```

Limitations. 303 patients from one centre in Iran; cross-validation is not external validation. Weaker for patients 65+ and for LCX and RCA. Indian patients often develop CAD younger, so the models need validating on Indian data before any clinical use. **Attribution:** dataset UCI #411 (Alizadehsani et al., CC BY 4.0); heart model “Realistic Human Heart” by neshallads (CC BY 4.0); EBM (MIT), CatBoost and XGBoost (Apache-2.0), LightGBM (MIT), scikit-learn (BSD), FastAPI, React, Three.js (MIT); Hindi voice by Sarvam AI. Full evidence: `reports/BEST_MODELS.md`, `models/manifest.json`.